

Multi-Store Demand Forecasting and Supply Chain Pipeline

Submitted by

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1. Abstract

Accurate demand forecasting is central to efficient retail supply chain management, particularly for organisations operating a large network of stores with heterogeneous sales patterns. This project develops a multi-store demand forecasting pipeline using a large-scale, real-world retail sales dataset comprising 421,570 weekly records across 45 stores and 81 departments spanning February 2010 to October 2012. The work begins with data cleaning, missing-value treatment and exploratory data analysis to understand seasonality, holiday effects, and the influence of store type and size on weekly sales. Time-aware features such as lag sales, rolling averages and rolling volatility were engineered to capture short-term demand momentum, alongside calendar features such as week, month, quarter and year. Four regression models — Linear Regression, Random Forest, Gradient Boosting and XGBoost — were trained on a chronological train-test split and evaluated using MAE, RMSE and R-squared. The Random Forest Regressor achieved the best performance with an R-squared of 0.9897, closely followed by XGBoost at 0.9892, both substantially outperforming the linear baseline. Feature-importance analysis showed that the 4-week moving average and recent lag sales were the dominant predictors, confirming strong short-term demand persistence. The resulting forecasts form the analytical core of a conceptual multi-store supply chain pipeline intended to support store-level replenishment and inventory planning decisions. The project demonstrates how a systematic data science workflow — from raw transactional data to a validated forecasting model — can be extended into an operational decision-support tool for retail supply chains.

2. Introduction

Retail organisations that operate through a large network of stores face a persistent operational challenge: predicting how much of each product will sell at each location, in each time period, so that inventory can be positioned correctly across the supply chain. Under-forecasting leads to stockouts and lost sales, while over-forecasting ties up working capital in excess inventory and increases holding costs. This challenge becomes more complex when demand is influenced by external and seasonal factors such as holidays, promotional markdowns, macroeconomic indicators (unemployment, consumer price index), fuel prices and local weather conditions, all of which vary across stores of different type and size. This project, titled “Multi-Store Demand Forecasting and Supply Chain Pipeline,” addresses this problem by building a data-driven forecasting pipeline that can be embedded into a retailer’s broader supply chain decision-making process.

The relevance of this project lies in its direct applicability to real-world retail operations: accurate, store-level demand forecasts are a prerequisite for effective replenishment planning, warehouse allocation, workforce scheduling and markdown strategy. From a technical standpoint, the project uses the Python data science stack — pandas and NumPy for data manipulation, Matplotlib and Seaborn for visualisation, and scikit-learn together with XGBoost for machine learning — executed in a Google Colab environment with data stored on Google Drive. Traditional retail forecasting historically relied on statistical time-series methods (moving averages, exponential smoothing, ARIMA); more recent industry and academic practice has shifted towards tree-based ensemble machine learning models, which can flexibly incorporate a large number of exogenous features (holidays, macroeconomic indicators, store attributes) alongside the historical sales signal, generally producing stronger accuracy on tabular retail datasets.

The overall procedure followed in this project moves through five stages: (i) data collection and understanding, (ii) data cleaning and preprocessing, (iii) exploratory data analysis to surface trends and relationships, (iv) time-aware feature engineering, and (v) model building, evaluation and selection. The purpose of the project is twofold: first, to produce a reliable weekly sales forecasting model at the store-department level; and second, to frame that forecasting capability as the analytical engine of a multi-store supply chain pipeline that can subsequently drive inventory replenishment and allocation decisions.

Training Topics Covered (First Two Weeks of Internship)

- Introduction

- Basic Statistics for Data Science
- Data Visualization
- Introduction to GitHub and Cloud Computing
- Data Science Application Development using Streamlit
- Machine Learning 1
- Machine Learning 2
- LLM Fundamentals
- Communication Skills
- Deep Learning 1

3. Project Objective

The project is guided by the following objectives:

1. To collect, clean and preprocess large-scale, multi-store, multi-department historical sales data so that it is suitable for reliable forecasting.
2. To perform exploratory data analysis to understand weekly sales trends, seasonality, holiday effects, and the influence of store type and store size on sales performance.
3. To engineer time-aware predictive features — including lag sales, rolling averages, rolling volatility and calendar attributes — that capture short-term and seasonal demand patterns.
4. To build, train and compare multiple regression models (Linear Regression, Random Forest, Gradient Boosting and XGBoost) for weekly sales forecasting, and to select the best-performing model using standard error and goodness-of-fit metrics.
5. To position the validated forecasting model as the analytical foundation of a multi-store supply chain pipeline that can inform inventory replenishment and allocation decisions.

No sample survey or hypothesis-testing exercise was conducted as part of this project; the analysis is based entirely on historical transactional and store-attribute data.

4. Methodology

4.1 Data Source and Description

The project uses a historical retail sales dataset (**Sales_Dataset.csv**) containing **421,570** weekly observations across **45** stores and **81** departments, covering the period from **5 February 2010** to **26 October 2012**. Each record represents the weekly sales for a given store-department combination, along with store attributes and external economic indicators. The key fields in the dataset are described below.

Field	Description
Store	Store identifier (1 to 45)
Dept	Department identifier within the store (1 to 81)
Date	Week-ending date of the sales record
Weekly_Sales	Total sales for the store-department in that week (target variable)
IsHoliday	Flag indicating whether the week contains a major holiday
Temperature	Average regional temperature for the week
Fuel_Price	Regional fuel price for the week
Markdown1–5	Promotional markdown values (anonymised, partially available)

Field	Description
CPI	Consumer Price Index for the region
Unemployment	Regional unemployment rate
Type	Store category (A, B or C)
Size	Physical size of the store

4.2 Data Cleaning and Preprocessing

The following steps were carried out to prepare the raw data for analysis and modelling:

1. Loaded the dataset and examined its shape, structure and summary statistics using `df.shape`, `df.info()` and `df.describe()`.
2. Checked for missing values: the `Markdown1`–`Markdown5` columns had substantial missingness (64% – 74%), while `CPI`, `Unemployment`, `Temperature` and `Fuel_Price` had no missing values in this pass.
3. Checked for and removed duplicate records (none were found in the dataset).
4. Imputed missing `Markdown` values with 0 (representing no active promotion) and imputed any missing `CPI`, `Unemployment`, `Temperature` and `Fuel_Price` values with their column median, a robust measure against outliers.
5. Converted the `Date` column to a proper datetime type to enable time-based operations.

4.3 Exploratory Data Analysis

With the cleaned dataset, exploratory analysis was performed to understand sales behaviour: the overall weekly sales trend across all stores, the difference in sales between holiday and non-holiday weeks, the top-performing stores by total sales, the relationship between store type/size and sales, the correlation between numeric features, and the average monthly sales pattern. The detailed charts and findings from this analysis are presented in Section 5.

4.4 Feature Engineering

To capture demand dynamics over time, the following engineered features were added at the store-department level, after sorting the data chronologically by `Store`, `Dept` and `Date`:

- Calendar features: `Year`, `Month`, `Week` (ISO week number) and `Quarter`, extracted from the `Date` column.
- Lag features: `Lag_1`, `Lag_2` and `Lag_4` — the weekly sales value 1, 2 and 4 weeks prior for the same store-department.
- Rolling features: a 4-week moving average (`MA_4`) and 4-week rolling standard deviation (`Rolling_STD_4`) of weekly sales, to capture short-term trend and volatility.
- Categorical encoding: the `Type` column (A/B/C) was label-encoded, and the boolean `IsHoliday` flag was converted to an integer (0/1).
- Rows containing NaN or infinite values introduced by the lag/rolling operations (at the start of each store-department series) were removed, and the final engineered dataset (408,436 rows × 25 columns) was saved for modelling.

4.5 Model Selection, Training and Validation

Because the data is a time series at the store-department level, a chronological (time-based) train-test split was used instead of a random split, to avoid information leakage from the future into the past and to realistically simulate a forecasting scenario. All records before 1 January 2012 were used for training, and all records from 1 January 2012 onwards were held out for testing.

Four regression models were trained on the engineered feature set and compared:

- Linear Regression — used as an interpretable baseline model.

- Random Forest Regressor (200 trees) — an ensemble of decision trees using bootstrap aggregation.
- Gradient Boosting Regressor (200 estimators, learning rate 0.05) — a sequential boosting ensemble.
- XGBoost Regressor (300 estimators, max depth 6, learning rate 0.05, subsample 0.8, colsample_bytree 0.8) — a regularised, optimised gradient-boosting implementation.

Model performance was evaluated using three complementary metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2), computed on the held-out test set. The best model was selected as the one achieving the lowest MAE/RMSE and the highest R^2 . Feature importance scores from the tree-based models were also examined to interpret which factors most strongly drive weekly sales.

4.6 Overall Pipeline

The end-to-end workflow followed in this project, from raw data to supply chain decision support, is summarised below.

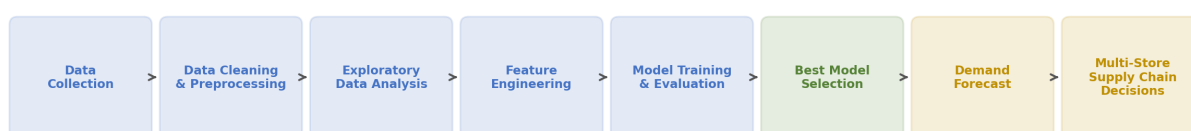


Figure 4.1: End-to-end multi-store demand forecasting and supply chain pipeline

All code developed for data cleaning, exploratory analysis, feature engineering and model training/evaluation has been organised in a Jupyter/Colab notebook and pushed to a GitHub repository: <https://github.com/sahithi0110/Walmart-Sales-Forecasting> (see Appendix C).

5. Data Analysis and Results

5.1 Dataset Overview

Metric	Value
Total records	421,570
Number of stores	45
Number of departments	81
Date range	05-Feb-2010 to 26-Oct-2012
Total cumulative weekly sales	₹67,372 crore (raw currency units: ~6.74 billion)
Duplicate records found	0
Records after feature engineering	408,436 rows × 25 columns

5.2 Missing Value Summary

Field	Missing (%)
Markdown1	64.26%
Markdown2	73.61%

Field	Missing (%)
MarkDown3	67.48%
MarkDown4	67.98%
MarkDown5	64.08%
All other fields	0%

As noted in the Methodology, MarkDown missingness was treated as “no promotion active” and imputed with 0.

5.3 Descriptive / Exploratory Analysis

Weekly Sales Trend: Aggregate weekly sales across all stores show a broadly stable pattern with sharp, recurring spikes in late November and December, corresponding to the Thanksgiving and Christmas holiday shopping periods in both 2010 and 2011.

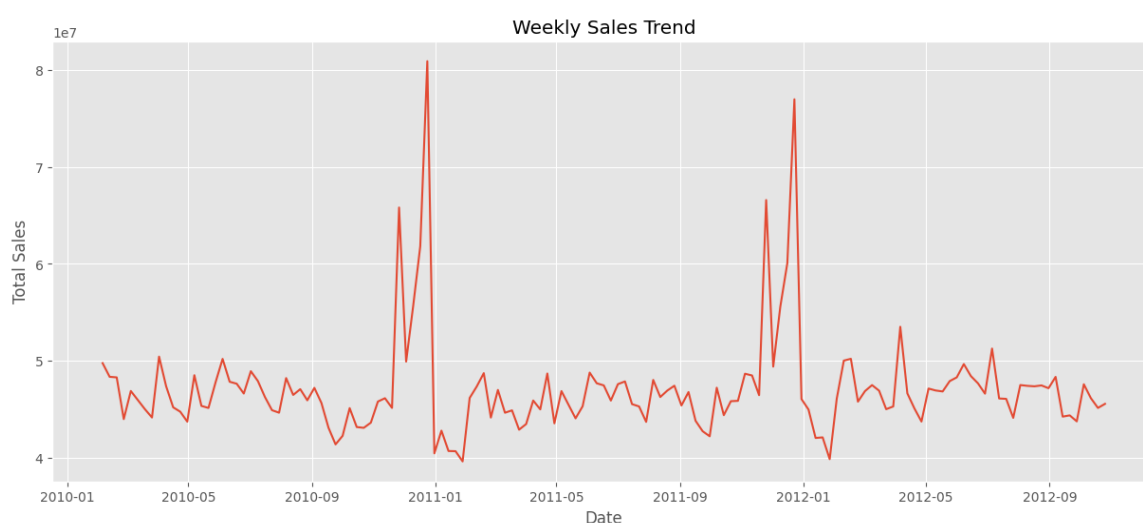


Figure 5.1: Aggregate weekly sales trend (2010–2012)

Holiday vs Non-Holiday Sales: Weekly sales during holiday weeks show a visibly higher median and a wider spread than non-holiday weeks, confirming that holidays are an important driver of short-term demand spikes that a forecasting model must capture.

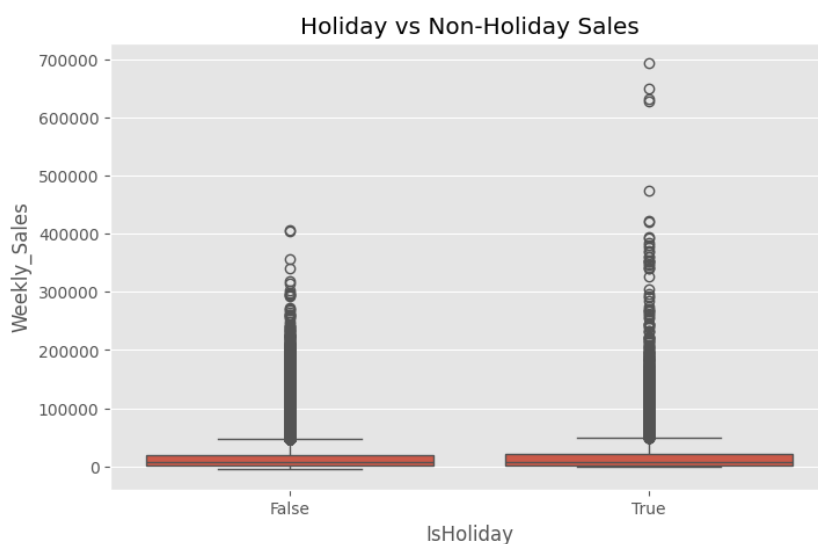
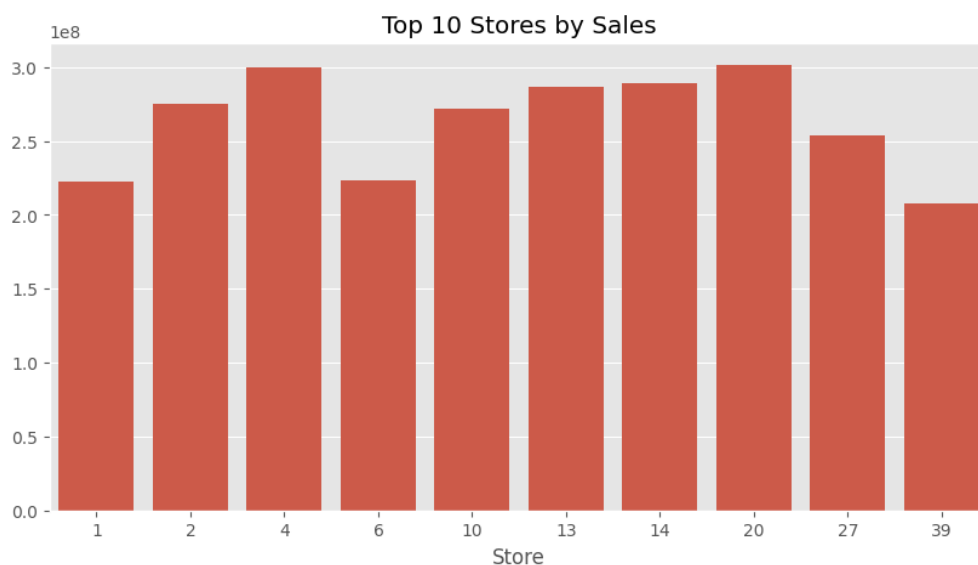
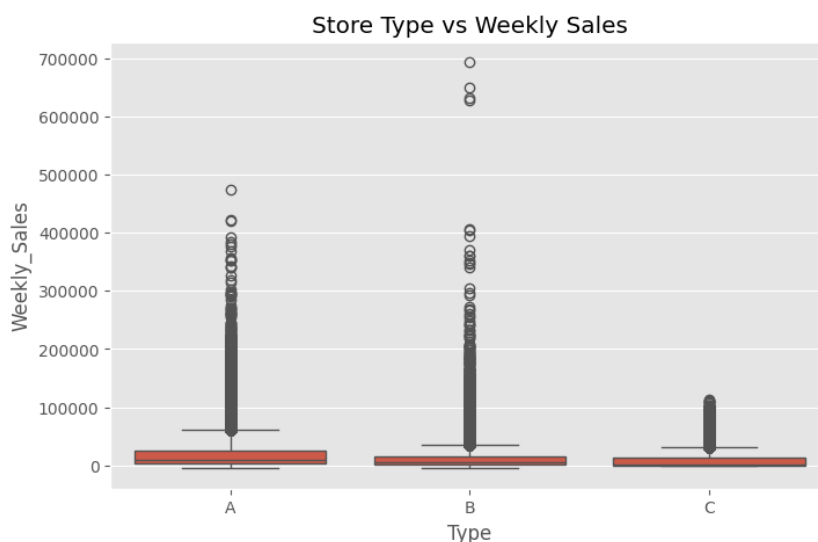


Figure 5.2: Distribution of weekly sales — holiday vs. non-holiday weeks

Top 10 Stores by Sales: A small number of stores contribute disproportionately to total sales, indicating that store-level heterogeneity is significant and should be reflected in the forecasting features (store ID, size, type).

*Figure 5.3: Top 10 stores ranked by cumulative sales*

Store Type vs Weekly Sales: Stores of Type A (generally the largest-format stores) show the highest median and most variable weekly sales, followed by Type B, with Type C stores showing the lowest and most consistent sales, aligning with their smaller footprint.

*Figure 5.4: Weekly sales distribution by store type*

Store Size vs Weekly Sales: A positive relationship is visible between store size and weekly sales — larger stores tend to generate higher and more variable sales — although the relationship is noisy, suggesting size alone is not sufficient to explain sales variation.

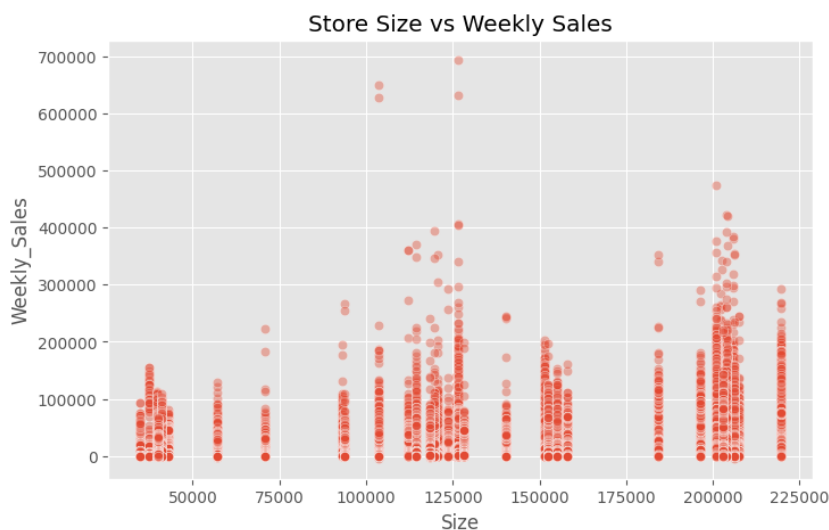


Figure 5.5: Store size versus weekly sales

Correlation Analysis: The correlation heatmap of numeric features shows that `Weekly_Sales` has its strongest positive association with `Size`, while most macroeconomic indicators (`CPI`, `Unemployment`, `Fuel_Price`, `Temperature`) show weak linear correlation with sales individually, motivating the use of non-linear, tree-based models that can capture interaction effects.

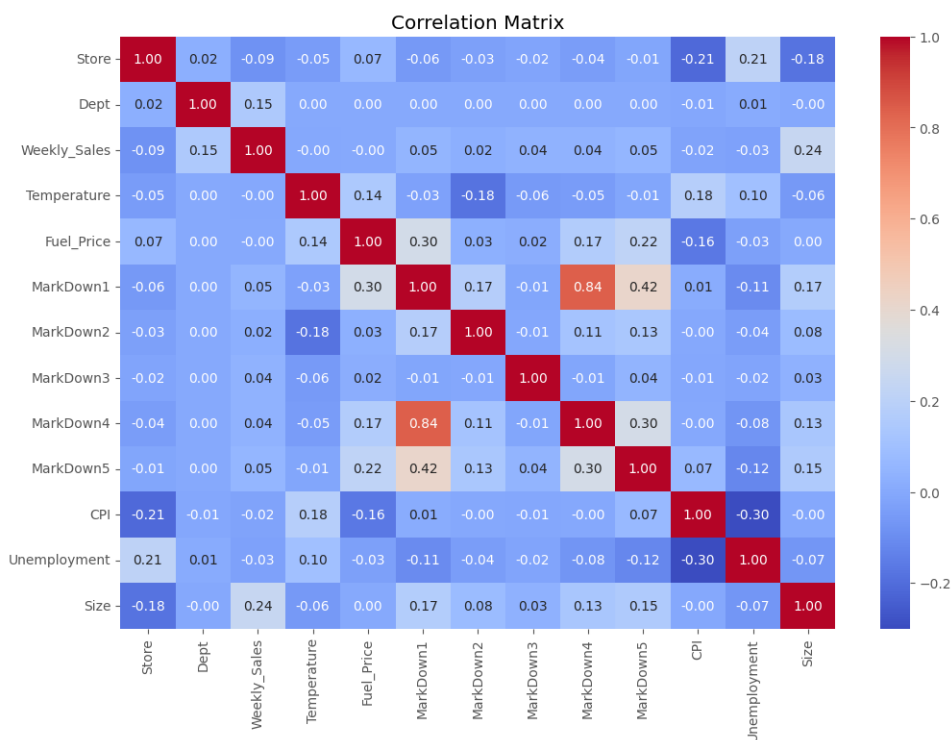


Figure 5.6: Correlation matrix of numeric features

Monthly Sales Trend: Average weekly sales peak around November–December (holiday season) and show a secondary, smaller lift around April, with a relative trough in January immediately following the holiday period.

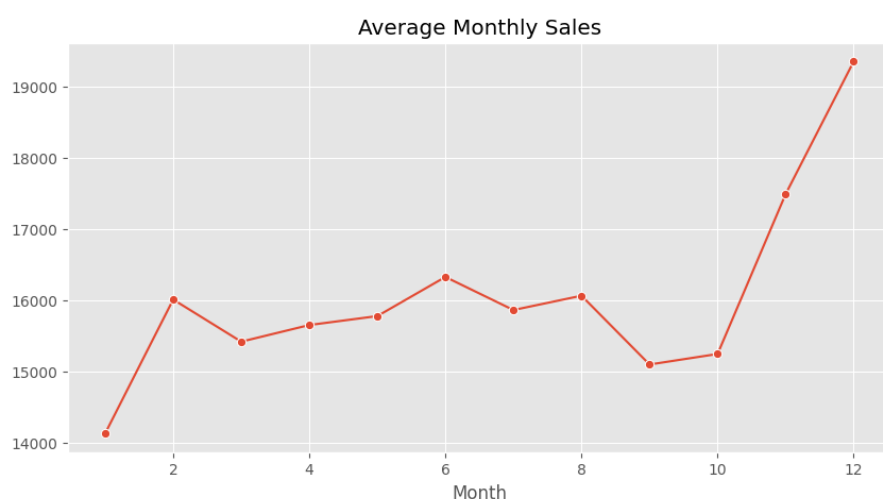


Figure 5.7: Average monthly sales trend

5.4 Model Comparison

All four models were evaluated on the chronologically held-out test set (records from January 2012 onwards). The results are summarised below, ranked by R^2 (best to weakest).

Model	MAE	RMSE	R^2
Random Forest	967.88	2,247.36	0.9897
XGBoost	1,091.42	2,300.48	0.9892
Gradient Boosting	1,372.23	2,909.68	0.9827
Linear Regression	1,401.25	3,103.59	0.9803

The Random Forest Regressor produced the lowest MAE and RMSE and the highest R^2 , making it the best-performing model in this comparison, closely followed by XGBoost. Both tree-based ensemble models comfortably outperformed the Linear Regression baseline, confirming that the relationship between the engineered features and weekly sales is meaningfully non-linear. All four models nonetheless achieved R^2 above 0.98, indicating that the lag and rolling-average features carry very strong predictive signal.

5.5 Actual vs. Predicted Sales

The plots below compare actual and predicted weekly sales for a sample of test-set observations, for the two best-performing models (Random Forest and XGBoost). In both cases the predicted series tracks the actual series closely, including around sharp demand spikes.

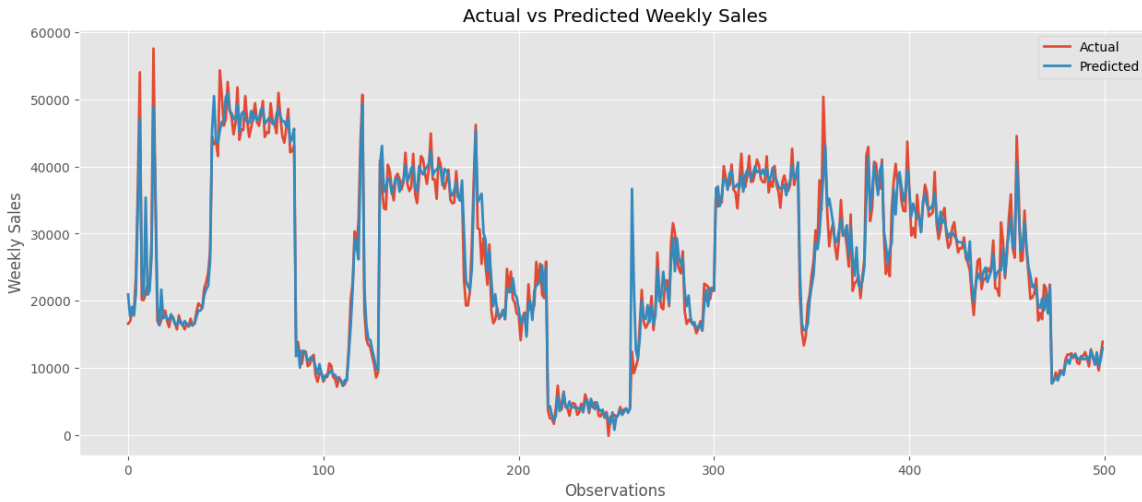


Figure 5.8: Actual vs. predicted weekly sales — Random Forest

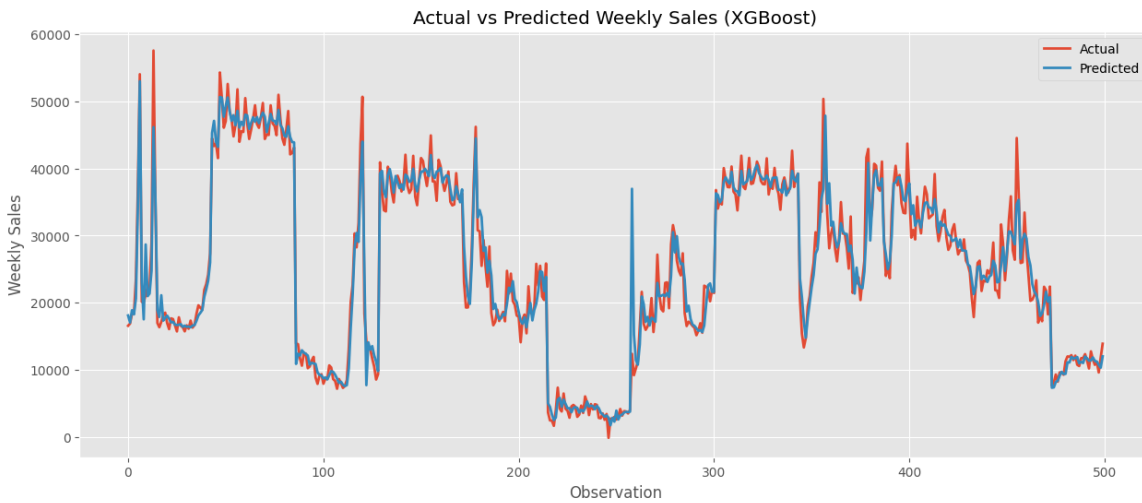


Figure 5.9: Actual vs. predicted weekly sales — XGBoost

5.6 Feature Importance

Feature importance scores from both tree-based models consistently identify the 4-week moving average (MA_4) as, by far, the most influential predictor of weekly sales, followed by the rolling standard deviation and the short-term lag features (Lag_1, Lag_2, Lag_4). Calendar features (Week, Month, Quarter, Year) and the IsHoliday flag contribute moderately, while macroeconomic indicators (CPI, Unemployment, Fuel_Price, Temperature) and markdown values contribute comparatively little on their own — consistent with the weak linear correlations observed in Section 5.3.

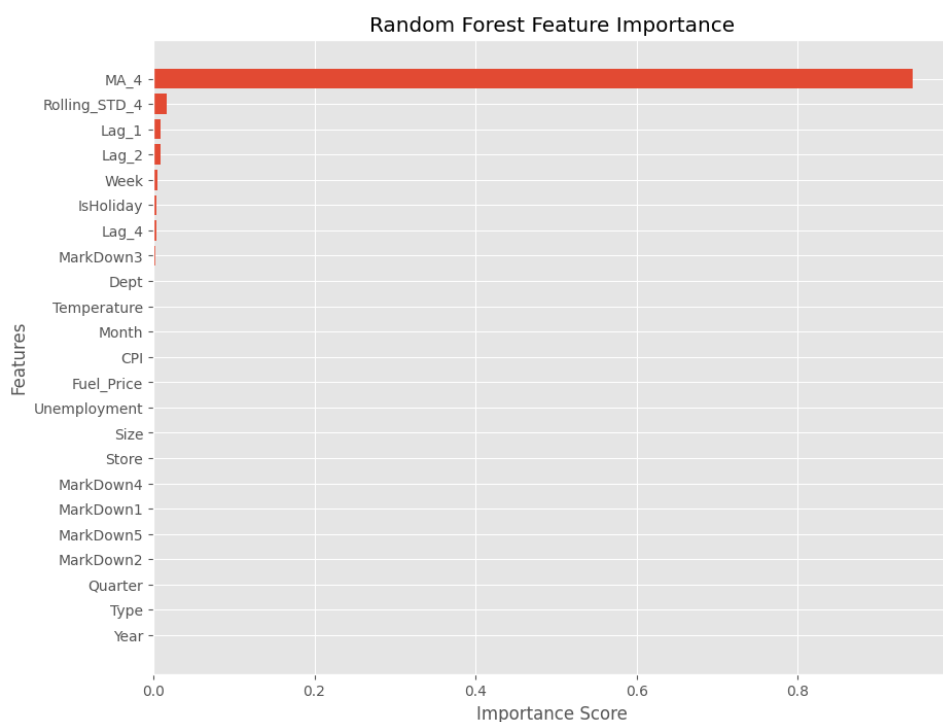


Figure 5.10: Random Forest feature importance

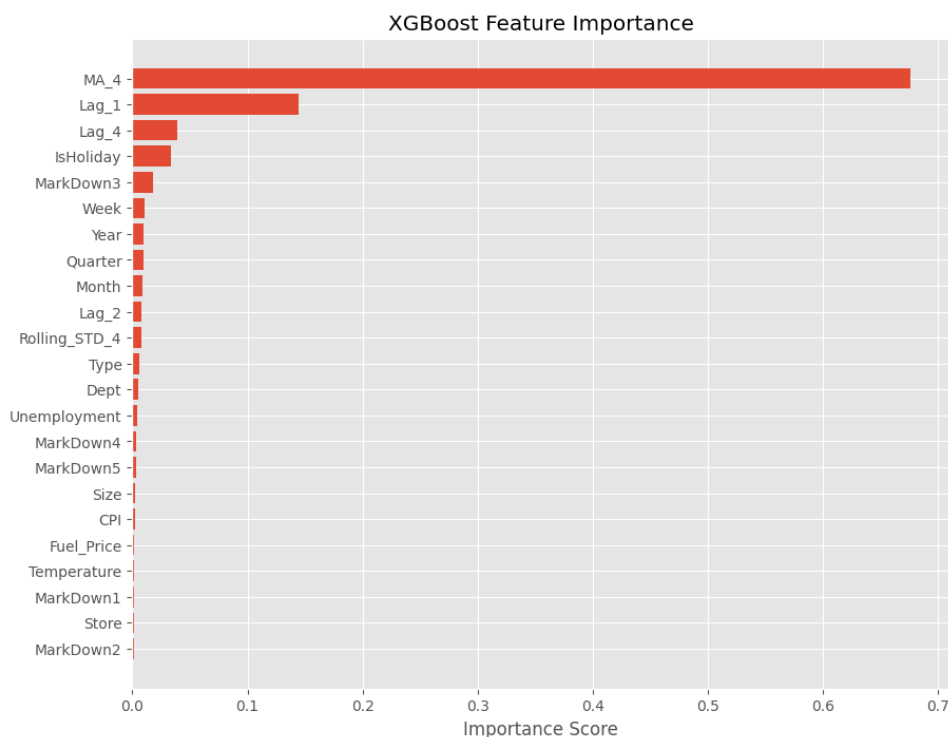


Figure 5.11: XGBoost feature importance

6. Conclusion

This project successfully developed and validated a multi-store weekly sales forecasting pipeline using a large, real-world retail dataset spanning 45 stores and 81 departments. Exploratory analysis confirmed the presence of strong seasonal and holiday-driven demand patterns, as well as meaningful differences in

sales behaviour across store type and size, justifying the inclusion of calendar, holiday and store-attribute features in the model.

Among the four models compared, the Random Forest Regressor delivered the best forecasting accuracy (MAE $\approx$ 967.9, RMSE $\approx$ 2,247.4, $R^2 \approx$ 0.99), narrowly ahead of XGBoost, and both substantially outperformed the Linear Regression baseline. Feature-importance analysis showed that recent sales history — captured through the 4-week moving average and short-term lag features — is the single strongest driver of near-term demand, more so than external economic indicators. This confirms that, for short-horizon retail forecasting, recency-based features carry the majority of the predictive signal, while holiday and calendar effects provide a valuable secondary boost.

From a supply chain standpoint, these results indicate that a Random Forest (or XGBoost) based forecasting model can serve as a dependable input to store-level replenishment planning — helping to anticipate holiday-driven demand surges, differentiate inventory allocation by store type and size, and reduce the risk of stockouts or overstock across the store network. The project therefore establishes the analytical foundation of a multi-store demand forecasting and supply chain decision-support pipeline.

Recommendations for future work include:

- Extending the model with dedicated time-series/deep-learning architectures (e.g., Prophet, LSTM, or Temporal Fusion Transformers) to better capture longer-horizon seasonality.
- Incorporating actual inventory, procurement lead-time and promotional calendar data to close the loop from forecast to a fully automated replenishment recommendation.
- Clustering stores by demand behaviour to enable cluster-specific forecasting models rather than a single global model.
- Deploying the selected model as a scheduled batch or real-time API service integrated with a supply chain planning system.
- Conducting formal hyperparameter tuning (e.g., grid/Bayesian search) and cross-validation to further improve model robustness.

7. Appendices

Appendix A: References

- Walmart Store Sales Forecasting dataset (retail weekly sales, store and economic-indicator data), as provided for this internship project.
- Pandas documentation — <https://pandas.pydata.org/docs/>
- Scikit-learn documentation — <https://scikit-learn.org/stable/documentation.html>
- XGBoost documentation — <https://xgboost.readthedocs.io/>
- Matplotlib / Seaborn documentation — <https://matplotlib.org/>, <https://seaborn.pydata.org/>

Appendix B: Survey Questionnaire

Not applicable — this project is based entirely on historical transactional data; no primary survey was conducted.

Appendix C: GitHub Repository

The complete source code (data cleaning, EDA, feature engineering, model training and evaluation) is available at the following repository link:

<https://github.com/sahithi0110/Walmart-Sales-Forecasting>

Appendix D: Other Document Links

A copy of this report, the underlying dataset, the trained model files and the project presentation are there in the shared GitHub repository link.